Compute Commons: Usable Access to Robot Inference

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Synthetic evidence · Final project · final-fp9-2026-10-21

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KEY MESSAGE

Winning compute does not guarantee usable service.
Fixed execution times matter: understating duration can raise utility by 8.

01 Question and foundations

Research question & verified gap

Can access policy deliver usable robot inference with truthful value bids?
Case: small integrators share a pick-and-place inference server.
RTC [1], Armory [2] and Themis [3] establish timing and allocation foundations.

Strategic model and Intellectual Merits

One sealed round; private task values, public state and eligibility.
Deadline = min(action-buffer time, pose-validity time) − network/guard delay.
Exact feasible allocation + critical payments gives truthful values under fixed public inputs.

Three complementary lenses

Game theory: utility requires actual timely delivery.
Social choice: entrant access can cost task value.
Mechanism design: change admission, payments and state verification together.

Auction comparison and boundary

Four jobs: values 10, 9, 8, 6; each 200 ms. Horizon 400 ms; deadlines 200, 200, 400, 400 ms.
Losers pay zero. Execution and settlement end the round. Local safety control is outside the auction.

02 Model and verified evidence

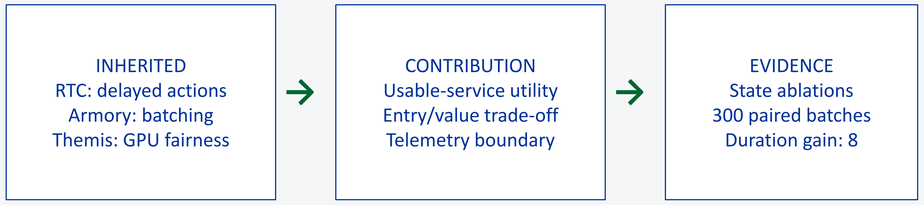


Figure 1. Inherited foundations → project contribution → synthetic evidence. No new general truthfulness theorem or Armory performance claim. References [1–3] below; full comparison in Appendix G.

Rule	Winners	Payments	Value	Revenue
Capacity only	A, B	8, 8	10	16
EDF	A, D	0, 0	18	0
Feasible	A, D	9, 6	18	15
Credit 3	A, E	9, 5	16	14
Reserve 100	A	9	10	9
Both: c=5	E	5	6	5

Capacity-only B pays 8 for a late result (utility −8). Credit trades 2 value units for entrant access.
Final test: 88,200 bid/deadline deviations, no positive gain; false duration raises utility 2 → 10.
36,289 original checks; 300 paired batches. Exact inputs, proof and limits: Appendix G.

03 Practical impacts and roadmap

Practical Impacts diagram with SDG targets

Verified admission + transparent credits → Operators / entrants Usable value + access

TARGET 4.4 · Technical learning
Indicator: prediction accuracy before/after explanation. Baseline: payment-only tutorial.
Risk: access loss / false safety inference. Safeguard: disclose trade-offs; retain local control. No measured human impact.

Future Research Roadmap · new diagram

Self-challenge → verify service duration, not just deadlines

ECONOMICS → Audit joint reports
Method: deviation grid; mentor: mechanism design
Criterion/milestone: bounded gain; tested audit rule

CS → Matched Armory workloads [2]
Data: robot traces; collaborator: serving researcher
Criterion/milestone: value, latency; baseline replay

BEHAVIOR → Compare explanations [4] · SDG 4.4
Data: consented pilot; FP10 if interested
Criterion/milestone: prediction error; preregistration

Integration and named contribution

Xinrong Sun: original access question; identified robot-specificity and telemetry challenges; directed manuscript, model, demo and poster revision. Reviewed earlier artifacts; final code checks recorded.
Economic utility changes both computation and prediction questions. No robot or human-study findings.

Feedback: Armory → narrow novelty; deadline challenge → test and refine verification boundary. Appendix D / G.

Full acknowledgments

Prof. Luyao Zhang for the course framework, project guidance and templates; Prof. Kenneth S. (Ken) Rogerson for feedback on improving the mechanism; and Han Zhang for sharing his project and urging us to distinguish robot inference from ordinary CPU allocation. Yiqiao Liu introduced Armory, challenged the novelty claim and proposed an equal-workload comparison through Ed and oral feedback; she also advised on referencing prior work in class. Xuantong Fu (FP3) identified the false-eligibility gain and called for reproducible, consistent artifacts. Muhan Chen provided oral and handwritten feedback. Tencent's Shanghai office provided the industry field-learning setting; the Shanghai Science and Technology Museum provided inspiration; and the COMSCI/ECON 206 classroom community enabled presentations and discussion. We credit Python, JavaScript, the course/ACM templates and cited research. Codex assisted revision and verification, as disclosed in Appendix A.

References · full entries and diagrams in Appendix G

[1] Kevin Black, Manuel Y. Galliker, Sergey Levine (2025). Real-Time Execution of Action Chunking Flow Policies. NeurIPS. doi:10.48550/arXiv.2506.07339
[2] Rohan Bansal, David He, Nadun Ranawaka Arachchige, Zhenyang Chen, Soobum Kim, Kexin Rong, Danfei Xu (2026). Action Chunk Scheduling for Batched Robot Policy Serving. doi:10.48550/arXiv.2608.00337
[3] Kshiteej Mahajan, Arjun Balasubramanian, Arjun Singhvi, Shivaram Venkataraman, Aditya Akella, Amar Phanishayee, Shuchi Chawla (2020). Themis: Fair and Efficient GPU Cluster Scheduling. NSDI, 289–304. usenix.org/conference/nsdi20/presentation/mahajan
[4] Hassidim, Marciano, Romm & Shorrer (2017). The Mechanism Is Truthful, Why Aren't You? AER 107(5), 220–224. doi:10.1257/aer.p20171027

OPEN MATERIALS · final-fp9-2026-10-21

github.com/dku-comsci-econ206-Autumn2026/
-PS2-FP9-ComputeCommons
huggingface.co/spaces/dku-comsci-econ206-2026/
compute-commons
Notebook + source hashes + response: repository
Official SDG source: sdgs.un.org/goals/goal4
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